East West University

Department of Computer Science and Engineering (CSE)
Semester: (Spring, Year: 2026), B.Sc. in CSE

Architectural Patterns Lab
Group: 2

Course Code: *CSE423*
Course Title: *Software Architecture*
Section: 01

Submitted to:

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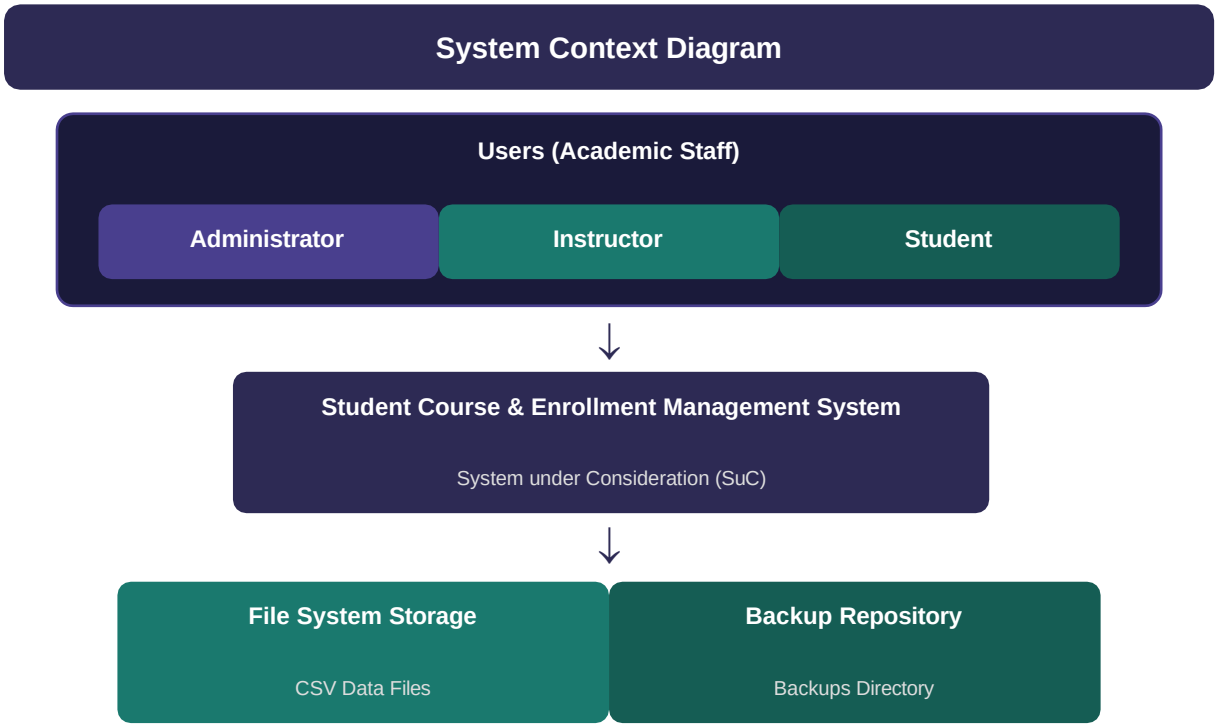
Submission Date: 1st May, 2026

Student Course & Enrollment Management System

Architecture Diagrams

1. Architectural Context Diagram with Archetypes

1.1 System Context Diagram



1.2 Archetype 1: Layered Architecture Archetype

Archetype 1: Layered Architecture Archetype

PRESENTATION ARCHETYPE

- CliManager — Main Application Entry Point
- Command Registry — Command Dispatcher
- Helpers: Student, Course, Enrollment, Data, Report Management

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BUSINESS LOGIC ARCHETYPE

- Services: StudentService, CourseService, EnrollmentService, ReportService, DataStore
- Design Patterns: Observer, Mediator, Decorator, Strategy, Bridge, Proxy

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DATA ACCESS ARCHETYPE

- ImportExportService, BackupService
- CsvServices (Student, Course, Enrollment)
- Strategy Pattern: ICsvImporter, ICsvExporter
- Adapter Pattern: ExportFormatAdapter

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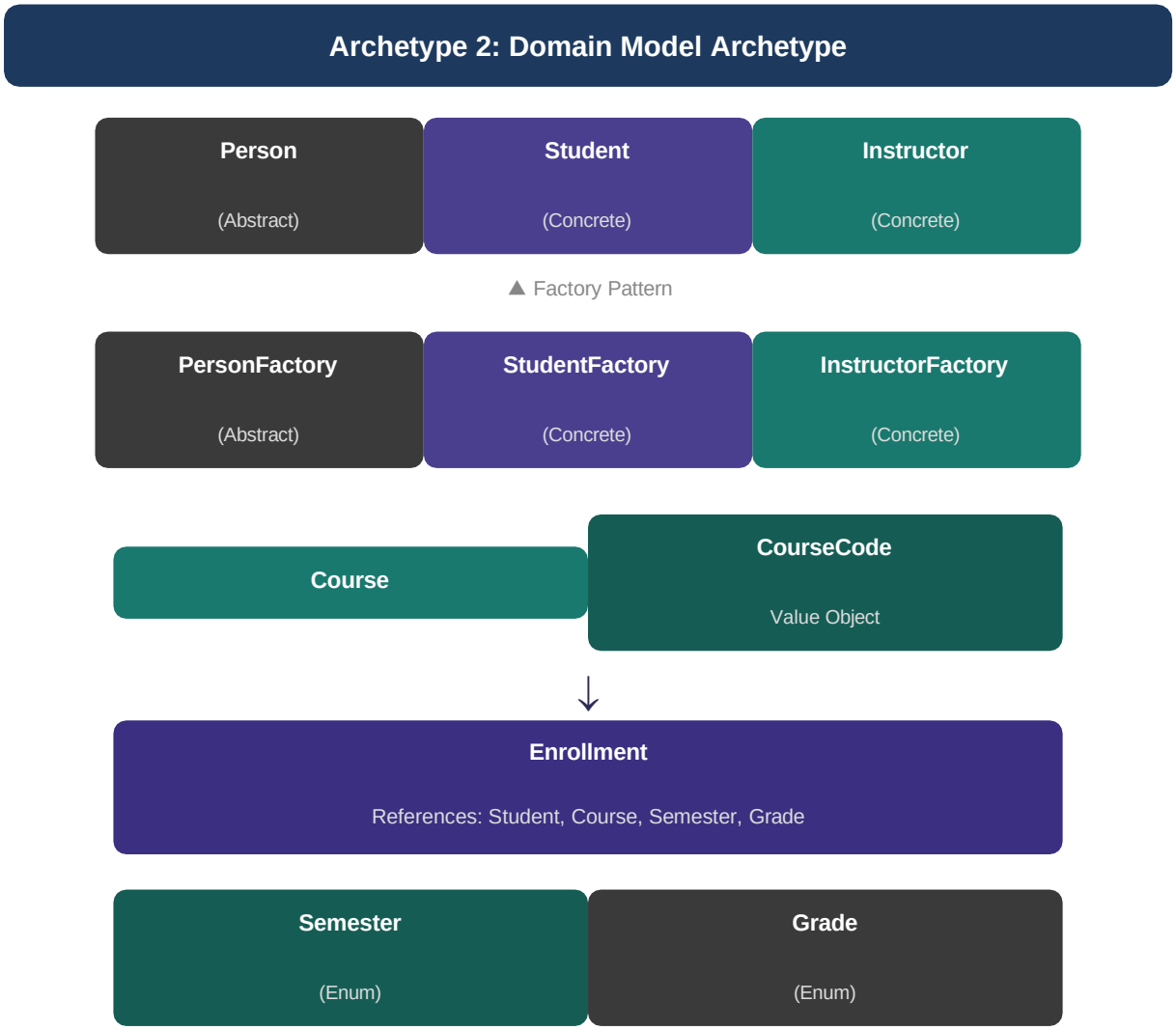
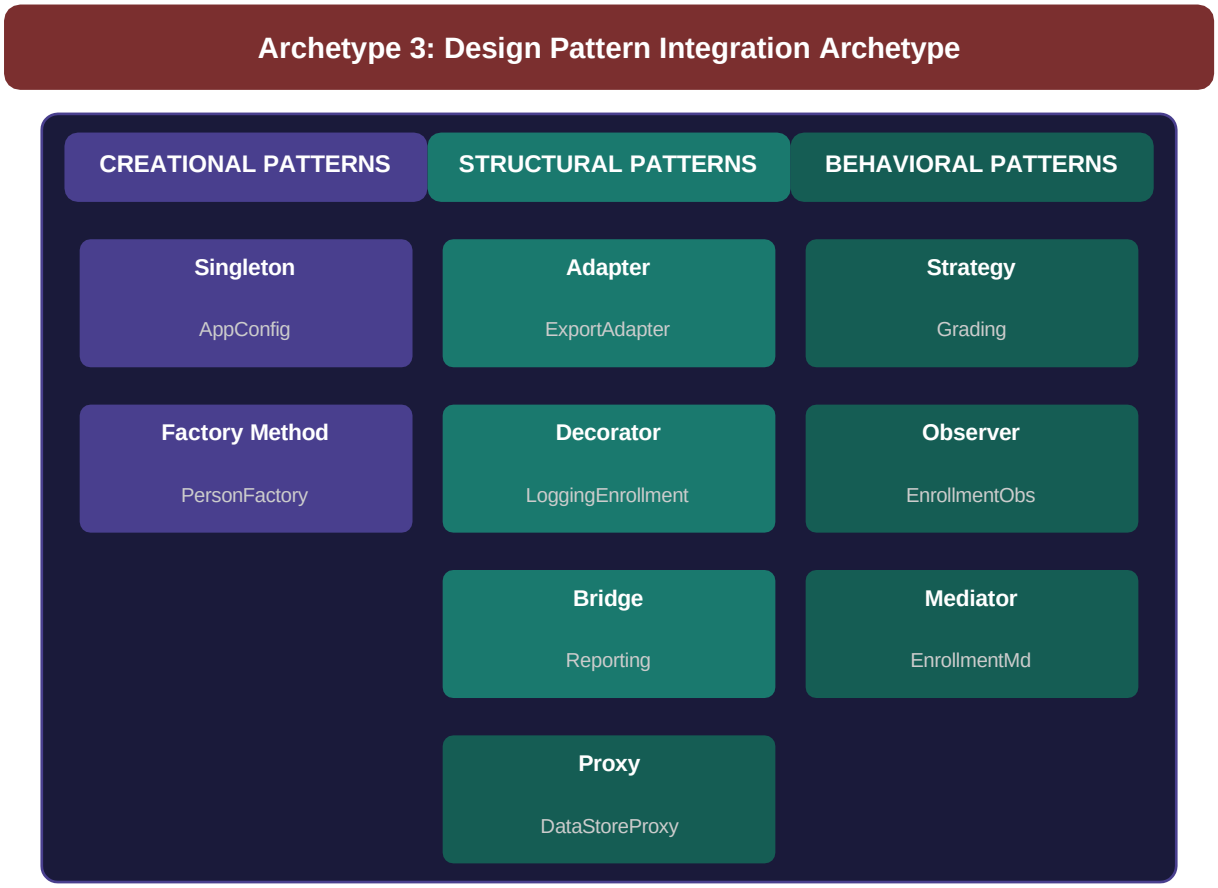
DATA STORAGE ARCHETYPE

- data/ : students.csv, courses.csv, enrollments_export.csv
- backup/ : timestamped backup folders

1.3 Archetype 2: Domain Model Archetype



1.4 Archetype 3: Design Pattern Integration Archetype



2. System Architecture Design — Layered Architecture Diagram

2.1 Complete Layered Architecture Diagram

Complete Layered Architecture Diagram

PRESENTATION LAYER

- CliManager.java — Main Application Controller
- CommandRegistry.java — Command Pattern Registry

helpers/

- StudentManagementHelper.java — Student UI Operations
- CourseManagementHelper.java — Course UI Operations
- EnrollmentManagementHelper.java — Enrollment UI Operations
- DataManagementHelper.java — Import/Export UI
- ReportManagementHelper.java — Report UI Operations

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BUSINESS LOGIC LAYER

Core Services:

- StudentService.java — Student Domain Operations
- CourseService.java — Course Domain Operations
- EnrollmentService.java — Enrollment Logic
- ReportService.java — Report Generation
- DataStore.java — In-Memory Data Repository

Design Patterns:

- mediator/ : ServiceMediator.java, EnrollmentMediator.java
- observer/ : EnrollmentObserver.java, EnrollmentNotifier.java, EnrollmentLogger.java
- decorator/ : EnrollmentServiceInterface.java, LoggingEnrollmentDecorator.java
- strategy/ : GradingStrategy.java, StandardGradingStrategy.java, RelativeGradingStrategy.java
- bridge/ : Report.java, GpaDistributionReport.java, ReportRenderer.java
- proxy/ : DataStoreInterface.java, DataStoreProxy.java

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DATA ACCESS LAYER

Data Services:

- ImportExportService.java — Import/Export Coordinator
- BackupService.java — Backup Management
- StudentCsvService.java — Student CSV Operations
- CourseCsvService.java — Course CSV Operations
- EnrollmentCsvService.java — Enrollment CSV Operations
- InstructorSetupService.java — Instructor Setup

Design Patterns:

- strategy/ : ICsvImporter.java, ICsvExporter.java
- StudentCsvImporter/Exporter, CourseCsvImporter/Exporter
- EnrollmentCsvExporter.java
- adapter/ : ExportFormatAdapter.java, CsvExportAdapter.java

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DOMAIN & UTILITIES LAYER

Domain Models:

- `Person.java (Abstract), Student.java, Instructor.java`
- `Course.java, CourseCode.java, Enrollment.java, Grade.java, Semester.java`

Factory Pattern:

- `PersonFactory.java (Abstract), StudentFactory.java, InstructorFactory.java`

Exceptions:

- `StudentNotFoundException, CourseNotFoundException`
- `DuplicateEnrollmentException, MaxCreditLimitExceededException`

Utilities:

- `InputValidator.java, GpaCalculator.java, TranscriptFormatter.java`
- `FileSystemUtils.java, RecursiveFileUtils.java`

Configuration:

- `AppConfig.java — Singleton Configuration`

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DATA STORAGE LAYER

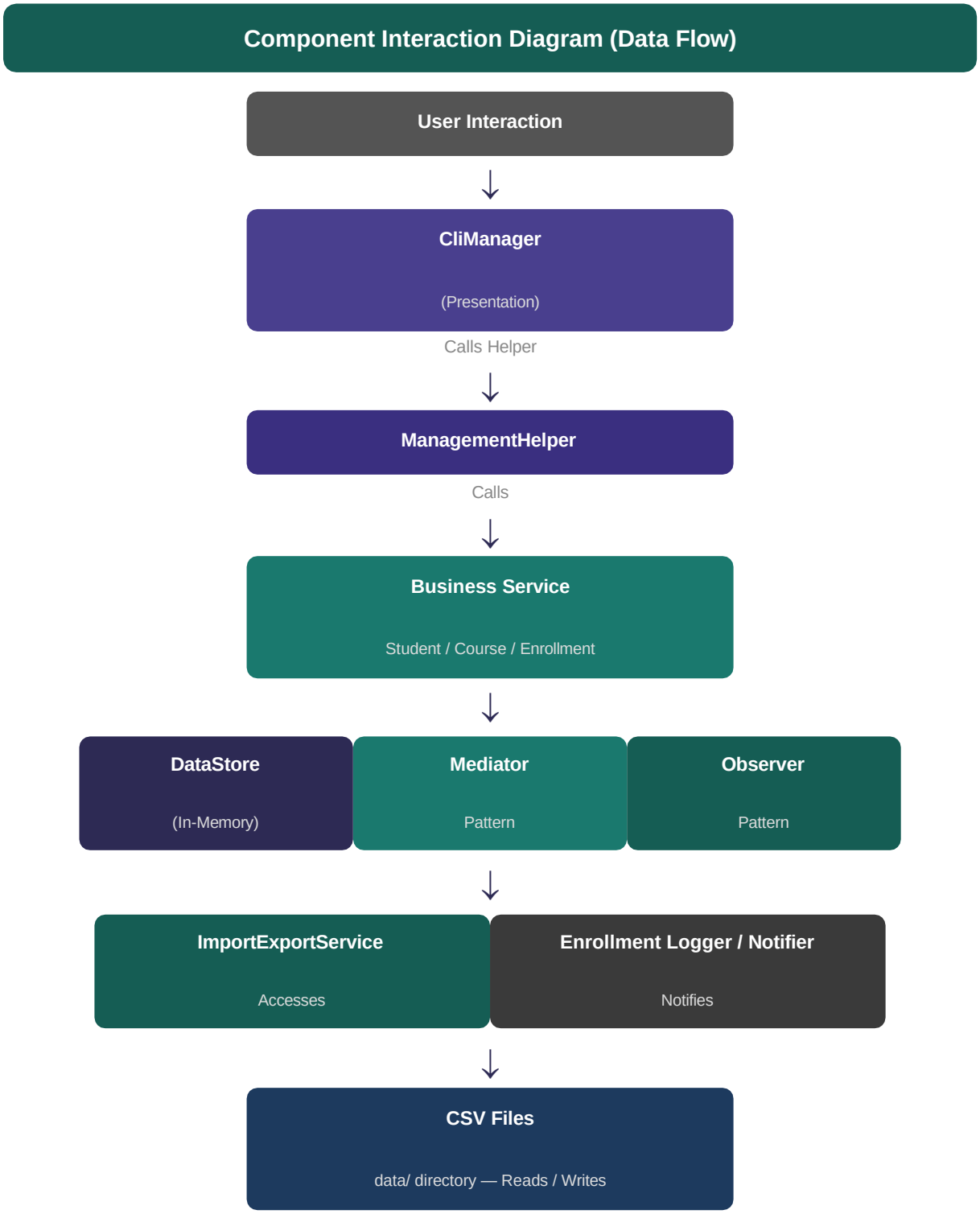
data/ Directory:

- `students.csv, courses.csv, students_export.csv`
- `courses_export.csv, enrollments_export.csv`

backup/ Directory:

- `backup_YYYY-MM-DD_HH-MM-SS/ — Timestamped Backups`

2.2 Component Interaction Diagram (Data Flow)



3. Event-Driven Architecture Diagram (Observer Pattern Detail)

Event-Driven Sub-Architecture (Observer Pattern Detail)

EVENT PRODUCER

EnrollmentService.java

```
addEnrollment() — Emits ENROLLMENT_CREATED event

removeEnrollment() — Emits ENROLLMENT_REMOVED event
```

Publishes Events

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EVENT CHANNEL (Observer List)

```
List<EnrollmentObserver> observers

Methods:

attach(EnrollmentObserver o)

detach(EnrollmentObserver o)

notifyObservers(EnrollmentEvent e)
```

Notifies All Observers

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Enrollment Notifier

(Console Output)

```
onEnrollmentCreated()

onEnrollmentRemoved()
```

Enrollment Logger

(Audit Log)

```
onEnrollmentCreated()

onEnrollmentRemoved()
```

Future Consumer

(Email/SMS Notifier)

```
onEnrollmentCreated()

onEnrollmentRemoved()
```